

Space Base Habitat Design

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Mission Statement

- The Harper Station Mars base is the first of its kind, a modular, expandable, and future-proof first of its kind settlement on the Isidis Planitia on the Red Planet.
- Here, groundbreaking scientific research on the history of the red planet is relayed back to Earth.
- At the same time, a new Martian colony images and develops itself.
- We will start with a 100 people, and develop a habitat capable of supporting 250+

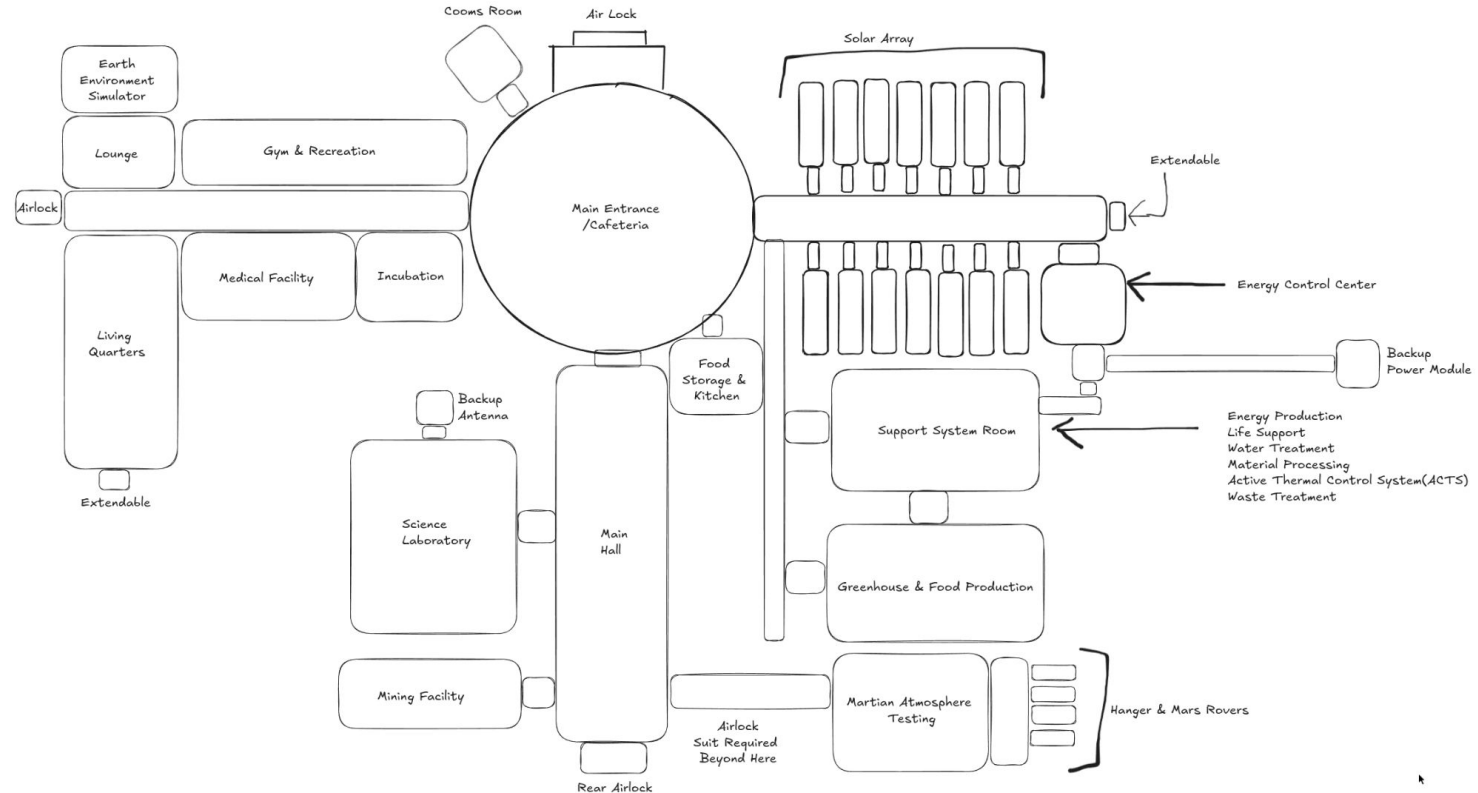
Crew Profile

- We will need a large variety of people to make this happen
- Engineers, Scientists, Physicists, Biologists Chemists, Electricians, Computer Scientists, Botanists, Doctors, Therapists, and Teachers will all work to make this mission a success.
- Due to the potential risk, these brave people will be aged 25-40 with no immediate family, ensuring minimal emotional deficit and able-bodiedness
- For the longevity of the base, the mission will be equipped with a diverse selection of fertilized eggs, capable of being grown into a large colony.

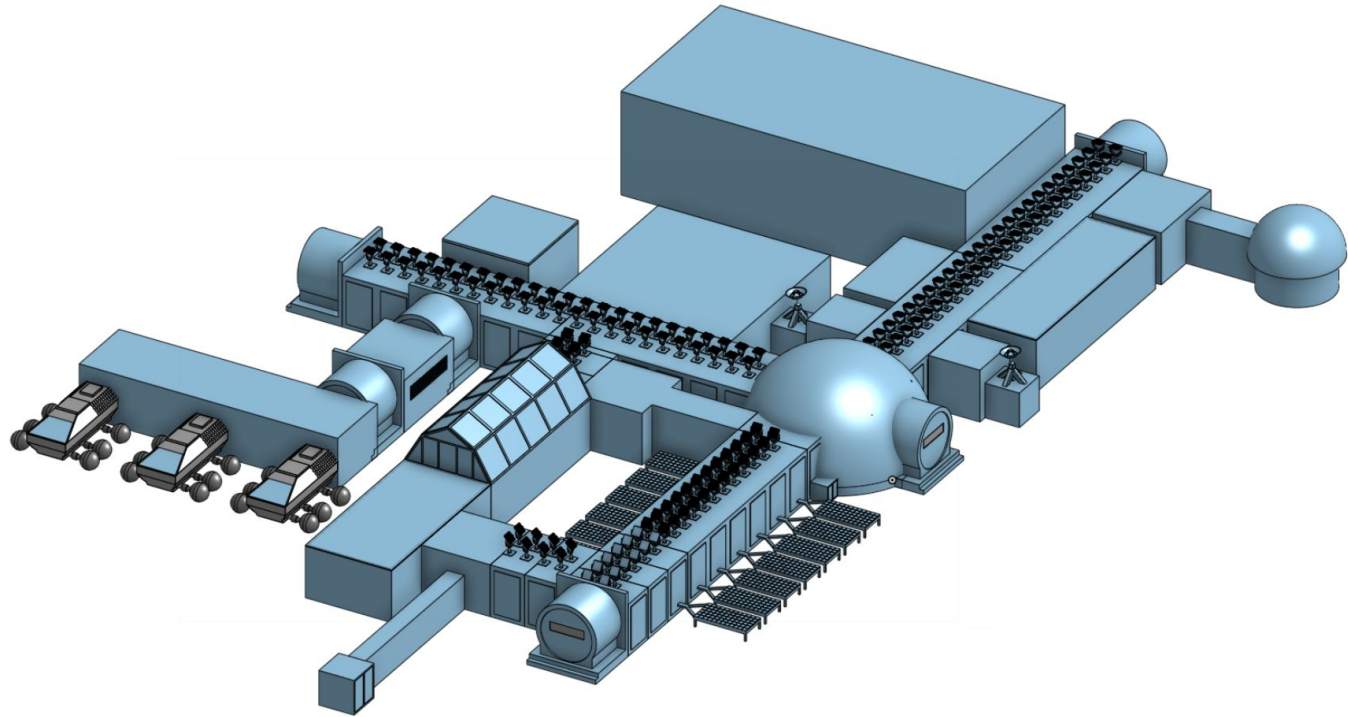
Expansion Strategy

- Once the initial crew has fully constructed a base capable of sustaining consistent food, water, air, and energy, we can allow for permanent settlement.
- We will then incubate the fertilized eggs, and while the children grow, we will expand the base to accommodate them.
- The design of the base is easily modular and expandable
- An unmanned station above the atmosphere will transit Earth payloads to Mars and allow for easy expansion.

Base Layout Diagram



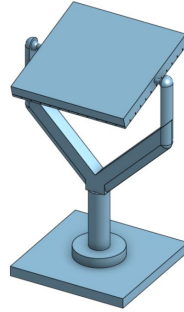
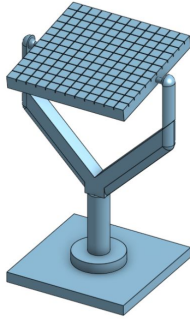
Base 3D Model



Materials Plan

- The very core of the base(the circular module) will start as the foundation. Due to its utmost importance, we will be producing it out of Titanium, a strong, lightweight metal.
- The rest of the base will have a frame constructed of aluminum, due to it being very light. This doesn't yield perfect structural rigidity, or radiation protection, however.
- To accommodate for this, we will utilize the ample sulfur available on the Martian Surface to create Sulfur Concrete to reinforce the aluminium frames in a number of ways.
- On the inside, we will line the interiors with aerogel, an super-effective insulator, in order to trap heat and maintain a steady internal temperature.

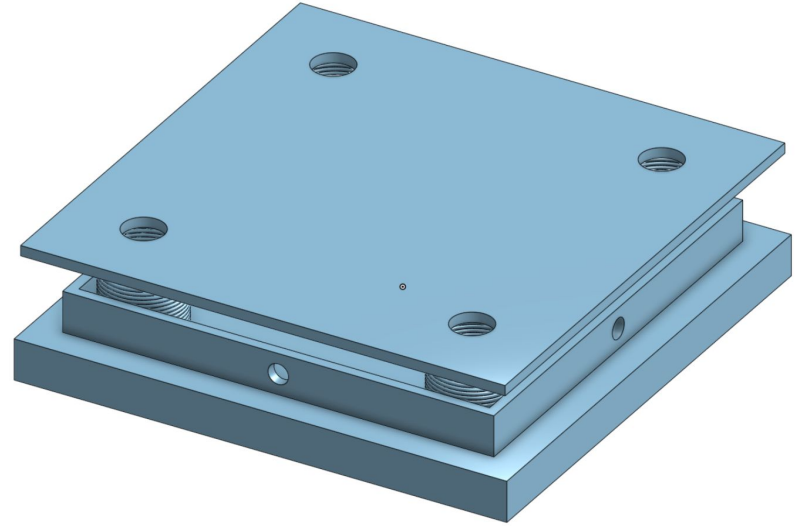
The Solar Shield



- The roof of core elements of the base will be outlined with the Solar Shield.
- This is a panel that on one side, acts a solar panel, providing electricity.
- But in the case of a sandstorm, meteor shower, or radiation storm, the Solar Shield can flip over, revealing a protective High-Density Polyethylene panel, providing incredibly strong protection to the base.

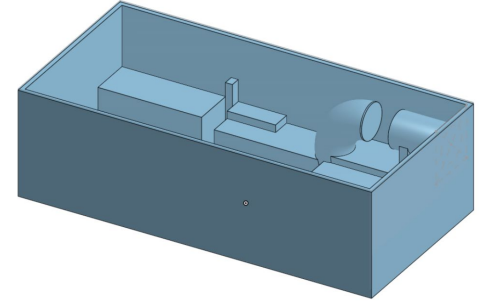
Internal Transportation Plan

- The current design of the base allows it to be very walkable
- We will not let this go waste, however, we will utilize piezoelectric flooring to harvest the kinetic energy of footsteps to power things like lights.



Atmosphere Plan

- We will generate the vast majority of our oxygen via the electrolysis of Water (Producing H_2 and O_2)
- Excess CO_2 will be used in tandem with H_2 to produce H_2O and Methane (CH_4) in a Sabatier Reactor.
- We will use an Active Thermal Control System (ACTS) to extract and re-distribute heat throughout the base.
- We will use an airlock-ventilation system with dehumidifiers to capture and re-distribute water to manage humidity,
- Air is circulated via pipes & vents and is recycled as much as possible.



Electrolysis of Water & Sabatier Reactor

The electrolysis of water is defined as follows: $2H_2O \rightarrow 2H_2 + O_2$, however, the E_{cell} of this electrolytic cell is <0 , and because $\Delta G = -nFE$, the Gibbs Free Energy of this reaction > 0 , meaning we need an external power source to power it, and drive the reaction reaction.

We can split up the reaction into the following:

Reduction: $2H^+(aq) + 2e^- \rightarrow H_2(g)$, $E = 0.00V$

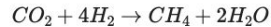
Oxidation: $2H_2O \rightarrow O_2 + 4H^+ + 4e^-$, $E = +1.23V$

$$E_{cell} = E_{cath} - E_{an}$$

$$= -1.23V$$

Thus we need a power source of 1.23 V or more to run our electrolysis

The Sabatier reaction is as follows:



Say we aim to produce 5kg of Methane per day, that is equal to

$$1.442 * 10^{-2} mol \frac{H_2}{s}$$

Then, using the equation: $I = \frac{dn}{dt} * 2F$, we get an amperage of 2783 Amps

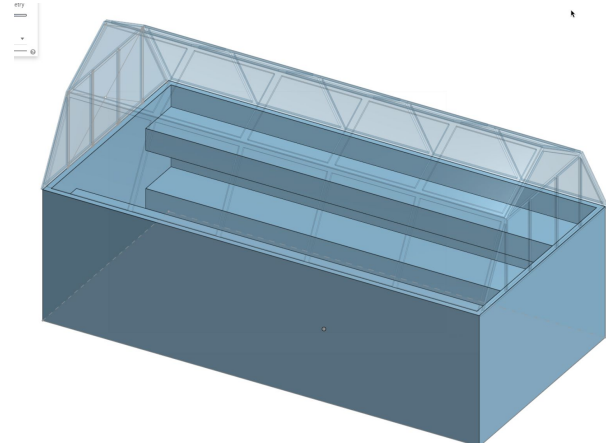
$P = IV$, $P=3.4Kw$, roughly 1.5 of these can come from using the produced methane in a natural gas generator, the other half will have to be provided from solar panels.

Water Management

- The Vast majority of our water can come from drilling the ample subsurface ice below the Martian Terrain.
- We can also re-collect moisture from the plants grown in the greenhouse
- The Sabatier reaction causes 5kg Methane a Day to be produced, 11.2 Liters of water are also produced a day from this reaction.
- We can use a water treatment machine to recycle human waste
- Water is stored in tanks, and we will need to mine quite a bit to provide 3.7 liters per person per day.
- In the future, we can expand the number of drills we have to mine more, and also expand the amount of Sabatier reactors we have to increase water production.

Food System Plan

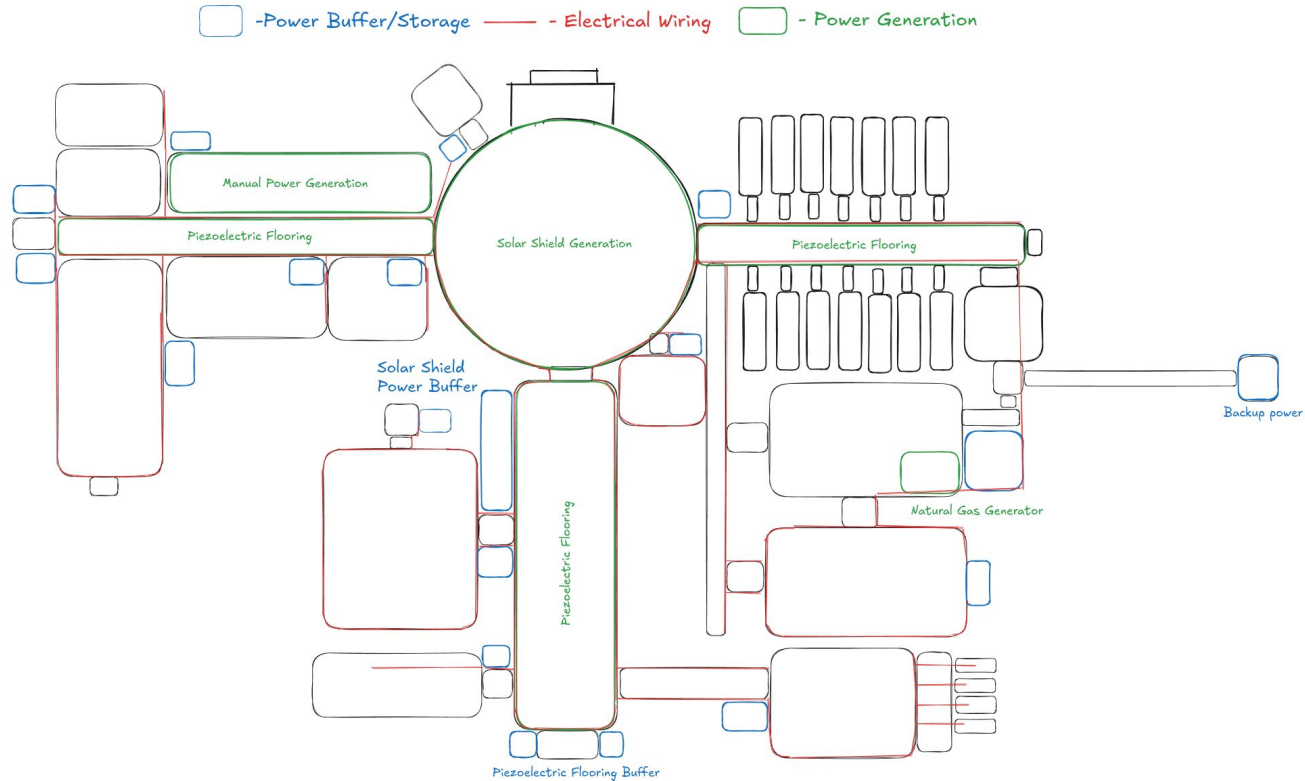
- We can mix Martian Regolith with both Salt and Fertilizer (derived from Human Excrement) to make arable soil.
- With this, we can grow crops like potato or lettuce in a greenhouse.
- For protein, we can grow beans, foster small insects like mealworms, or utilize lab grown meat.
- We will also boast a kitchen, where people can make their own food
- Nearby we will have warm and cold food storage
- Various combinations of these vegetables and protein options can lead to a widely varied diet.



Power Source Plan

- There were a number of available power sources, however, good-old Solar Panels seem to do the best job.
- Nuclear power, while more efficient, is incredibly risky in such a foreign environment, and leads to waste, and possible contaminations.
- At the same time, the base is not large enough, or tall enough for that matter, to benefit from any large scale wind energy. The location of Isidis Planitia does not provide any geothermal energy, so that rules out another option.
- Isidis Planitia is, however, located on the equator, and gets the most sun out of any possible landing spot on Mars, making solar a great option.
- Although solar is fully reliable, with a wide enough array we can power our base.
- This becomes even more sustainable when you mix in a natural gas generator burning methane generated from a Sabatier reactor and energy generated from humans either from workout machine powered turbines or piezoelectric flooring

Energy Distribution Diagram



Energy Distribution Plan

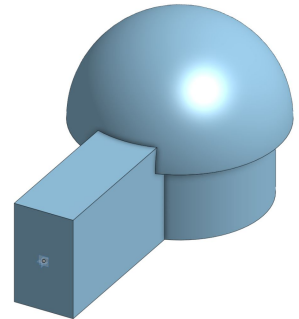
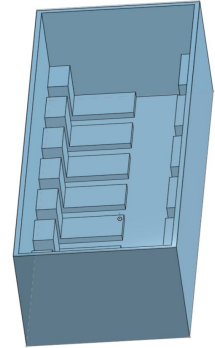
- We will have a central power hub where all energy is generated
- Here, we will house the main capacitor, which will distribute power throughout the base as needed
- Each module has its own buffer capacitor as a redundancy
- The Exercise Generator, Solar Shield Generator, and Piezoelectric Flooring will all also each store power in their own respective capacitors.

Emergency Backup Power

- We will also house an emergency backup power capacitor, with a physical cutoff switch separated from the main grid
- This backup capacitor can power the entire base for 2 days
- Because it is physically disconnected, it requires manual enabling, so it is truly a last failsafe.

Physical & Mental Health Plan

- The gym area provides a variety of rowing machines, cables, and treadmills allowing exercise, reduction of inhabitants muscle atrophy, and the added bonus of power generation via spun turbines.
- The medical bay is equipped with multiple physicians, and a deep care room
- The medical bay is also equipped with a therapist.
- For mental health, the lounge carries a large variety of Earth media to keep people from feeling homesick
- At the same time, there is also a dedicated Earth simulator room
- This room will contain a catwalk inside of a sphere of 360 degree monitors
- It will also contain wind blowers, mist sprayers, and 7.1 surround sound audio
- With this, inhabitants can simulate different Earth-like environments and minimize homesickness.

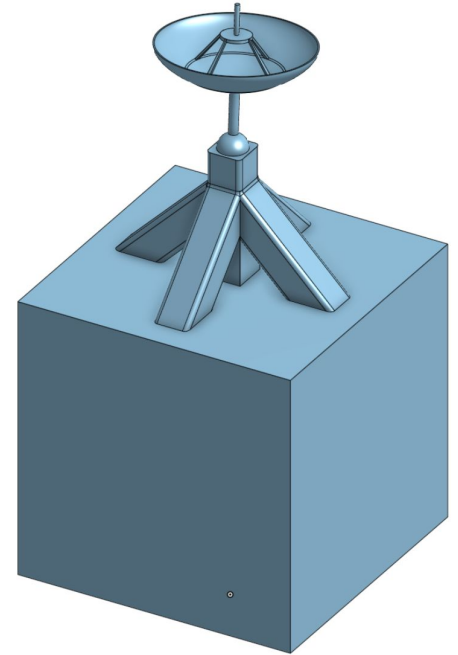


Living Arrangements

- The quarters are modular, and can easily be built & added onto
- Each room is relatively spacious and isolated, equipped with ample storage for personal items.
- There is a multi-toilet and multi-shower restroom every 5 quarters, equipped with needed amenities.
- The Kitchen and cafeteria are available in the main building.
- On a day-to-day basis, people with either live in their room, the gym, the lounge, or be working on science in the lab
- Privacy is easily available via living quarters, but there is a feeling of shared community about the whole base

Earth Communication Plan

- We have a main communications room with a large antenna capable of sending both text, audio, and video.
- Here, people can message and send videos to loved ones, as well as relay scientific findings back to Houston
- People can access the communications api from any one of their computers in the lab; going into this room is not required
- Data is related to the aforementioned unmanned space station, which will amplify the signal towards earth, ensuring maximum communication speed and coverage,
- Data from earth is transmitted to the space station, which then send the info down to the base via its downlink
- There is also a second antenna located at the other end of the base for redundancy.



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